

Vision

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 - a- 0.01
 - b- 0.10
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2. The intra-ocular pressure:
 - a- Is the pressure of vitreous humor
 - b- Increases the power of lens
 - c- If increased produces cataract
 - d- If increased prevents near vision
 - e- If decreased produces glaucoma
3. Glaucoma:
 - a. Is a disease which causes opacity of lens
 - b. Is a disease which affects far vision
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4. Glaucoma:
 - a. Of the open angle type is due to forward movement of iris
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5. In the corneal reflex:
 - a- The afferent fibers run in the optic nerve
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7. Concerning the near response:
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9. In order for accommodation to occur:
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 - b- The shape of the lens must be changed
 - c- The ciliary muscle is relaxed for near vision
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 - e- None of these answers
10. An increase in refractive power is produced by contraction of the:
 - a. Iris
 - b. Ciliary muscle
 - c. Suspensory ligaments
 - d. Extraocular muscles
 - e. Pupil
11. When emmetropic persons become presbyopic, their:
 - a. Visual acuity increases
 - b. Near point increases
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 - d. Total refractive power increases
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12. In hypermetropia:
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13. Total loss of accommodation in persons over 70 years of age is called:
 - a- Amblyopia
 - b- Emmetropia
 - c- Hypermetropia
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14. A 58-year-old woman goes to her physician because she is having difficulty threading a needle. After physical examination she is diagnosed with presbyopia (old eyes). Her condition is caused by which of the following?
- A. Clouding of the vitreous.
 - B. Retinal detachment
 - C. Ciliary muscle paralysis
 - D. Stiffening of the lens
 - E. Degeneration of the macula
15. Accommodation for far vision requires:
- a- Constriction of the pupil of the eye
 - b- Dilation of the pupil of the eye
 - c- An increase in the formation of rhodopsin
 - d- An increase in the curvature of lens
16. Which of the following statements concerning myopia is correct?
- a- The curvature of the lens is uneven
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 - c- A convex lens is used to correct the condition
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17. Regarding astigmatism:
- a- Light rays emitted from different planes do not come to a common focal point
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18. Regarding the power of the eye:
- a- Hypermetropia may be corrected by a diverging lens
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- a- The intraocular pressure is about 1.5 mmHg
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20. The circular muscle of the iris:
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 - b- Is innervated by the parasympathetic nervous system
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 - e- All of the above
21. One of the following occurs with constriction of the pupil:
- a- Increased production of aqueous humor
 - b- Sympathetic nerve fibers are stimulated
 - c- There is increased depth of focus
 - d- Vision in conditions of poor illumination is improved
 - e- Chromatic aberration is increased
22. Concerning the pupillary light reflex:
- a- The center lies in the superior colliculus
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23. The light reflex involves the following structures, except:
- a. Edinger-Westphal nucleus
 - b. Ciliary ganglion
 - c. lateral geniculate body
 - d. Oculomotor nerve
24. All about meiosis of the pupil is correct, except:
- a. Reduces visual field
 - b. Increases depth of field
 - c. Increases the visual acuity
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25. Concerning the eye all the following is correct, except:
- a- The fovea lies on the visual axis
 - b- Aqueous humour is secreted into the anterior chamber
 - c- Light reflexly activates the parasympathetic nerves supplying smooth muscle in iris
 - d- Most of the focusing power is due to the cornea

26. In what layer of the retina is a large store of vitamin A found?
- a. The choroid
 - b. The outer nuclear layer
 - c. The pigmented layer
 - d. The outer plexiform layer
 - e. The inner nuclear layer
27. Melanin in the pigmented layer of retina:
- a- Is precursor of the light sensitive rhodopsin
 - b- Serves as nutritional component for rods and cones
 - c- Is a dark pigment that prevent reflection of light inside the eye
 - d- Maintains integrity of canal of schlemm
28. Visual contrast is enhanced due to lateral inhibition by which of retinal cells?
- a- Amacrine cells
 - b- Bipolar cells
 - c- Ganglion cells
 - d- Horizontal cells
 - e- Photoreceptors
29. The fovea contains an increased proportion of:
- a- Cones
 - b- Ganglion cells
 - c- Horizontal cells
 - d- Rods
 - e- Vasculature
30. The blind spot is:
- a- Located 5 degrees lateral to the central point of vision
 - b- Exit point of optic nerve
 - c- Contains only rods and thus has monochromatic vision
 - d- Contains no blood vessels
 - e- Area where chromatic aberration of the lens is greatest
31. Rods:
- a. Are sensitive to high intensity light
 - b. Are mainly concentrated at the fovea
 - c. Are primarily involved in color vision
 - d. Have longer receptor potential than cones
 - e. Are more sensitive to red light

32. The fovea centralis:

- a. Is crossed by blood vessels
- b. Contains a greater abundance of rods than cones.
- c. Is the point on the retina at which the optic nerve leaves the eye
- d. Is responsible for scotopic vision
- e. Lies on the temporal side of the optic disc

33. The fovea of the eye

- a- Contains only rods
- b- Is situated over the exit of the optic nerve
- c- Is the region of highest visual acuity
- d- is the thickest part of the retina

34. Cones:

- a. are most abundant in the periphery of the retina.
- b. are more numerous than rods.
- c. have higher sensitivity to light than rods.
- d. are responsible for color vision.
- e. contain pigments which are most affected by blue green light.
- f. have longer receptor potential than rods

35. When rods are stimulated, the following occurs:

- a. Light is necessary for the conversion of all-trans retinal to 11-cis retinal
- b. cGMP levels increase
- c. Rods depolarize
- d. Metarhodopsin II activates G protein
- e- Increased Na^+ conductance of the outer segment of the receptors
- f- The release of chemical transmitter increases

36. Rhodopsin:

- a- Is the photopigment found in the red cones
- b- Consists of an opsin and a chromophore part
- c- Is most highly concentrated in the fovea
- d- Is slowly broken down in the absence of light
- e- Contains a derivative of vitamin B₁₂

37. When rhodopsin is decomposed by light energy, what happens to change the membrane potential of the rod?

- a- The activity of the sodium pump at the inner segment is increased
- b- Membrane permeability in the outer segment is increased
- c- The membrane permeability for potassium greatly increases
- d- Membrane permeability for sodium ions in the outer segment is decreased

38. After stimulation of photoreceptors, which of the following occurs?
- a- Phosphodiesterase activity decreases
 - b- Transducin activity decreases
 - c- Hydrolysis of cGMP increases
 - d- Neurotransmitter release increases
 - e- Number of open voltage-gated calcium channels increases
39. Concerning the retinal photopigments:
- a. They are of 3 types
 - b. The opsin (protein part) is the same in all photopigments
 - c. Light is necessary for the conversion of all-trans retinal to 11-cis retinal
 - d- The photopigment rhodopsin is highly sensitive to red light
 - e. They are regenerated in the dark
40. Which of the reactions in the retinal rods is caused directly by the absorption of light energy?
- A- Dissociation of scotopsin and metarhodopsin
 - b- Dissociation of scotopsin
 - C- Transformation of 11- cis retinal to all - trans retinal
 - d- Transformation of metarhodopsin to lumirhodopsin
 - e- Transformation of vitamin A to retinene
41. When rods are stimulated, the following occurs:
- A. cGMP levels increase.
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 - D. Increased Na conductance of the segment of the receptors.
42. Stimulation of photoreceptors by light produces which of the following:
- a. Phosphodiesterase activity decreases
 - b. Formation of metarhodopsin II increases
 - c. Neurotransmitter release increases
 - d. Increased Na⁺ conductance of the outer segment of the receptors
 - e. Activation of rhodopsin kinase enzyme occurs
43. Isomerization to all-trans retinal after absorption of light:
- a- Inactivates guanylyl cyclase
 - b- Activates transducin
 - c- Occurs in rods but not cones
 - d- Results in opening of cGMP-activated channels
 - e- Inactivates phosphodiesterase

44. Which of the following pairs of molecules combine to form rhodopsin?
- a- Bathorhodopsin and 11-cis retinal
 - b- Bathorhodopsin and all-trans retinal
 - c- Bathorhodopsin and scotopsin
 - d- Scotopsin and 11-cis retinal
 - e- Scotopsin and all-trans retinal
45. Regarding retina:
- a- Total number of cones in retina is much greater than total number of rods
 - b- Each cone responds to all wave lengths of light
 - c- Photoreceptor activation results in hyperpolarization of the receptor
 - d- Pigment layer of retina contains the photoreceptors
46. Vitamin A deficiency causes:
- a- Poor night vision
 - b- Poor color vision
 - c- Astigmatism
 - d- Presbyopia
 - e- Myopia
47. All about photoreceptors is correct, except:
- a- Contain pigments which are broken down, or bleached by light
 - b- Synapse directly with ganglion cells in the retina
 - c- Generate hyperpolarizing potentials in response to light
 - d- In fovea are all cones
48. Which of the following retinal cells has action potentials?
- a. Bipolar cells
 - b. Horizontal cells
 - c. Photoreceptors
 - d. Ganglion cells
49. The Ganglion cells:
- a. Of magno type are involved in color vision
 - b. Of parvo type respond to changes in light intensity
 - c. Fire spontaneously in the dark
 - d. Which have off center-on surround receptive field are inhibited when light stimulates the receptors in surround
50. Concerning color vision:
- a. The ability to recognize yellow-green depends on rhodopsin
 - b. a person is dichromates when he has a weak cone system
 - c. Green is perceived when only green cones are stimulated
 - d. The stimulation ratio of the 3 types of cones allows specific color perception

- e- When there is no stimulation of red, green or blue cones there is sensation of seeing white
- f- A person with protanopia can see green color properly

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52. Regarding vision:

- a- Color vision depends the 4 retinal photopigments
- b- Full color vision is possible in dim light
- c- Full dark adaptation takes nearly 30 minutes
- d- In a dark-adapted eye, the visual acuity is best at the center of the visual field

53. Regarding visual cortex:

- a- Simple cells respond best to moving bars without change in their directions
- b- Visual agnosia is due to lesion of area 17
- c- Blobs are concerned with color coding
- d- The visual cortex of one side receives fibers from the nasal side of the ipsilateral retina

54. Cutting which structure on the right side causes blindness in the temporal field of the left eye and the nasal field of the right eye?

- a- Optic nerve
- b- Optic chiasma
- c- Optic tract
- d- Primary visual cortex
- e- Secondary visual cortex

55. Which of the following conditions is caused by a lesion of the optic chiasma?

- a- Bitemporal hemianopia
- b- Blindness of both eyes
- c- Homonymous hemianopia
- d- Heteronymous hemianopia
- e- Scotoma

56. Which one of the following visual lesions will produce extensive loss of peripheral vision in the left visual fields of both eyes?
- a. A lesion of the left lateral geniculate
 - b. Section of the left optic nerve
 - c. A lesion of the right visual cortex
 - d. Removal of the right lens
 - e. Section of the left optic tract
57. The visual acuity is:
- a- A measure of sensitivity of retina to light
 - b- Greater in a person with 6/12 vision than in one with 6/9
 - c- Greater using central than using peripheral vision
 - d- Depends on the stimulation of 2 adjacent cones
58. Which of the following is correct?
- a- In Argyl Robertson pupil accommodation reflex is lost and pupillary reflex is intact
 - b- Astigmatism occurs because the lens elasticity is decreased.
 - c- Visual aphasia Occurs due to lesion in the cortical areas 18 and 19.
 - d- Hypermetropia is corrected by biconcave lenses.

10- In myopia:

- a- The curvature of the cornea is unequal in different planes
- b- A far object is focused on the retina by using accommodation
- c- The near point is less than 10 cm but near vision is not impaired
- d- The images from the two eyes are not fused within the cortex
- e- There is increased intraocular pressure due to abnormal long eyeball

12- Stimulation of photoreceptors by light produces:

- a- Increased activity of phosphodiesterase by activated transducin
- b- Light-dependent conversion of all-trans retinal to 11-cis retinal
- c- Depolarization with increased release of neurotransmitter
- d- Increased activity of the Na^+ pump at the inner segment
- e- Increased levels of cGMP in the outer segment

10- Pigmented layer of the retina:

- a- Contains cells that produce lateral inhibition of photoreceptors
- b- Is represented in the electroretinogram by the [b] wave
- c- Is highly developed at the periphery of retina
- d- Is the main site for storage of vitamin A

11- A 16-year-old boy enters a darkened movie theater on a sunny summer afternoon. It takes many minutes for his eyes to adapt to the darkness because:

- a- It takes some time for the retina to rebuild its rhodopsin store
- b- Rods in eye work only in bright light, and it takes time for cones to work in weak light
- c- The iris takes time to close after being subjected to bright light.
- d- The optic nerve fatigues in bright light, and it takes time to recover

12- The Ganglion cells:

- a- Responds to stimulation by a local state of depolarization
- b- Of the parvo- type show transient discharge and are very responsive to movements
- c- Of on-center off-surround receptive field have receptors with excitatory transmitter
- d- Are very excitable cells which can fire spontaneously even inhibited by dark current

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 - c- Bathorhodopsin and scotopsin
 - d- Scotopsin and 11-cis retinal**
 - e- Scotopsin and all-trans retinal
45. Regarding retina:
- a- Total number of cones in retina is much greater than total number of rods
 - b- Each cone responds to all wave lengths of light
 - c- Photoreceptor activation results in hyperpolarization of the receptor**
 - d- Pigment layer of retina contains the photoreceptors
46. Vitamin A deficiency causes:
- a- Poor night vision**
 - b- Poor color vision
 - c- Astigmatism
 - d- Presbyopia
 - e- Myopia
47. All about photoreceptors is correct, except:
- a- Contain pigments which are broken down, or bleached by light
 - b- Synapse directly with ganglion cells in the retina**
 - c- Generate hyperpolarizing potentials in response to light
 - d- In fovea are all cones
48. Which of the following retinal cells has action potentials?
- a. Bipolar cells
 - b. Horizontal cells
 - c. Photoreceptors
 - d. Ganglion cells**

49. The Ganglion cells:
- a. Of magno type are involved in color vision
 - b. Of parvo type respond to changes in light intensity
 - c. Fire spontaneously in the dark**
 - d. Which have off center-surround receptive field are inhibited when light stimulates the receptors in surround
50. Concerning color vision:
- a. The ability to recognize yellow-green depends on rhodopsin
 - b. a person is dichromatic when he has a weak cone system
 - c. Green is perceived when only green cones are stimulated
 - d. The stimulation ratio of the 3 types of cones allows specific color perception**
 - e- When there is no stimulation of red, green or blue cones there is sensation of seeing white
 - f- A person with protanopia can see green color properly
51. Concerning color vision:
- A. Green is perceived when only green cones are stimulated.
 - B. the stimulation ratio of the 3 types of cones allows specific color perception.
 - C. The stimulation ratio of the 3 types of cones allows specific color perception.**
 - D. When there is no stimulation of red, green, or blue cones there is sensation of Seeing white.
 - E. A person with protanopia can see green color properly.
52. Regarding vision:
- a- Color vision depends on the 4 retinal photopigments
 - b- Full color vision is possible in dim light
 - c- Full dark adaptation takes nearly 30 minutes**
 - d- In a dark-adapted eye, the visual acuity is best at the center of the visual field
53. Regarding visual cortex:
- a- Simple cells respond best to moving bars without change in their directions
 - b- Visual agnosia is due to lesion of area 17
 - c- Blobs are concerned with color coding**
 - d- The visual cortex of one side receives fibers from the nasal side of the ipsilateral retina

54. Cutting which structure on the right side causes blindness in the temporal field of the left eye and the nasal field of the right eye?
- a- Optic nerve
 - b- Optic chiasma
 - c- Optic tract**
 - d- Primary visual cortex
 - e- Secondary visual cortex
55. Which of the following conditions is caused by a lesion of the optic chiasma?
- a- Bitemporal hemianopia**
 - b- Blindness of both eyes
 - c- Homonymous hemianopia
 - d- Heteronymous hemianopia
 - e- Scotoma
56. Which one of the following visual lesions will produce extensive loss of peripheral vision in the left visual fields of both eyes?
- a. A lesion of the left lateral geniculate
 - b. Section of the left optic nerve
 - c. A lesion of the right visual cortex**
 - d. Removal of the right lens
 - e. Section of the left optic tract
57. The visual acuity is:
- a- A measure of sensitivity of retina to light
 - b- Greater in a person with 6/12 vision than in one with 6/9
 - c- Greater using central than using peripheral vision**
 - d- Depends on the stimulation of 2 adjacent cones
58. Which of the following is correct?
- a- In Argyl Robertson pupil accommodation reflex is lost and pupillary reflex is intact
 - b- Astigmatism occurs because the lens elasticity is decreased.
 - c- Visual aphasia Occurs due to lesion in the cortical areas 18 and 19.**
 - d- Hypermetropia is corrected by biconcave lenses.

10- In myopia:

- a- The curvature of the cornea is unequal in different planes
 - b- A far object is focused on the retina by using accommodation
 - c- The near point is less than 10 cm but near vision is not impaired
 - d- The images from the two eyes are not fused within the cortex
 - e- There is increased intraocular pressure due to abnormal long eyeball
-

12- Stimulation of photoreceptors by light produces:

- a- Increased activity of phosphodiesterase by activated transducin
 - b- Light-dependent conversion of all-trans retinal to 11-cis retinal
 - c- Depolarization with increased release of neurotransmitter
 - d- Increased activity of the Na⁺ pump at the inner segment
 - e- Increased levels of cGMP in the outer segment
-

10- Pigmented layer of the retina:

- a- Contains cells that produce lateral inhibition of photoreceptors
- b- Is represented in the electroretinogram by the [b] wave
- c- Is highly developed at the periphery of retina
- d- Is the main site for storage of vitamin A

11- A 16-year-old boy enters a darkened movie theater on a sunny summer afternoon. It takes many minutes for his eyes to adapt to the darkness because:

- a- It takes some time for the retina to rebuild its rhodopsin store
- b- Rods in eye work only in bright light, and it takes time for cones to work in weak light
- c- The iris takes time to close after being subjected to bright light.
- d- The optic nerve fatigues in bright light, and it takes time to recover

12- The Ganglion cells:

- a- Responds to stimulation by a local state of depolarization
 - b- Of the parvo- type show transient discharge and are very responsive to movements
 - c- Of on-center off-surround receptive field have receptors with excitatory transmitter
 - d- Are very excitable cells which can fire spontaneously even inhibited by dark current
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